

Ibn Tofail University

Analysis V

Problem Set IV

Exercise 1:

For $(x, y) \in \mathbb{R}^2$, we define

$$f(x, y) = \begin{cases} xy \sin\left(\frac{\pi}{2} \cdot \frac{x^2 - y^2}{x^2 + y^2}\right) & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

1. Study the continuity of f .
2. Study the differentiability and determine if f is of class C^1 .
3. Study the second-order partial derivatives and determine if f is of class C^2 .

Correction

1. Continuity:

Let's examine the limit as $(x, y) \rightarrow (0, 0)$. Using polar coordinates $x = r \cos \theta$, $y = r \sin \theta$:

$$\begin{aligned} f(r \cos \theta, r \sin \theta) &= r^2 \cos \theta \sin \theta \sin\left(\frac{\pi}{2} \cdot \frac{r^2 \cos^2 \theta - r^2 \sin^2 \theta}{r^2}\right) \\ &= \frac{r^2}{2} \sin(2\theta) \sin\left(\frac{\pi}{2}(\cos^2 \theta - \sin^2 \theta)\right) \end{aligned}$$

Since $|\sin(\cdot)| \leq 1$ and $|\sin(2\theta)| \leq 1$, we have:

$$|f(r \cos \theta, r \sin \theta)| \leq \frac{r^2}{2} \rightarrow 0 \quad \text{as } r \rightarrow 0$$

Therefore, $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0 = f(0, 0)$, so f is continuous at $(0, 0)$.

2. Differentiability and class C^1 :

Let's compute partial derivatives at $(0, 0)$:

$$\begin{aligned} \frac{\partial f}{\partial x}(0, 0) &= \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0 \\ \frac{\partial f}{\partial y}(0, 0) &= \lim_{k \rightarrow 0} \frac{f(0, k) - f(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{0 - 0}{k} = 0 \end{aligned}$$

Now let's check if f is differentiable at $(0, 0)$:

$$\lim_{(h,k) \rightarrow (0,0)} \frac{f(h,k) - f(0,0) - 0 \cdot h - 0 \cdot k}{\sqrt{h^2 + k^2}} = \lim_{(h,k) \rightarrow (0,0)} \frac{hk \sin\left(\frac{\pi}{2} \cdot \frac{h^2 - k^2}{h^2 + k^2}\right)}{\sqrt{h^2 + k^2}}$$

In polar coordinates:

$$= \lim_{r \rightarrow 0} \frac{r^2 \cos \theta \sin \theta \sin\left(\frac{\pi}{2}(\cos^2 \theta - \sin^2 \theta)\right)}{r} = \lim_{r \rightarrow 0} r \cos \theta \sin \theta \sin\left(\frac{\pi}{2}(\cos^2 \theta - \sin^2 \theta)\right) = 0$$

So f is differentiable at $(0,0)$.

For $(x,y) \neq (0,0)$, the partial derivatives exist and are continuous. Therefore f is of class C^1 on $\mathbb{R}^2$.

3. Second-order partial derivatives and class C^2 :

Let's compute second partial derivatives at $(0,0)$:

$$\frac{\partial^2 f}{\partial x^2}(0,0) = \lim_{h \rightarrow 0} \frac{\frac{\partial f}{\partial x}(h,0) - \frac{\partial f}{\partial x}(0,0)}{h}$$

For $(x,y) \neq (0,0)$:

$$\frac{\partial f}{\partial x}(x,y) = y \sin\left(\frac{\pi}{2} \cdot \frac{x^2 - y^2}{x^2 + y^2}\right) + xy \cos\left(\frac{\pi}{2} \cdot \frac{x^2 - y^2}{x^2 + y^2}\right) \cdot \frac{\pi}{2} \cdot \frac{2x(x^2 + y^2) - 2x(x^2 - y^2)}{(x^2 + y^2)^2}$$

This expression is complicated near $(0,0)$. Let's check along different paths:

Along $y = 0$: $\frac{\partial f}{\partial x}(x,0) = 0$, so $\frac{\partial^2 f}{\partial x^2}(0,0) = 0$

Along $x = y$: $\frac{\partial f}{\partial x}(x,x) = x \sin(0) + x^2 \cos(0) \cdot \frac{\pi}{2} \cdot \frac{2x(2x^2) - 2x(0)}{4x^4} = x^2 \cdot \frac{\pi}{2} \cdot \frac{4x^3}{4x^4} = \frac{\pi}{2}$

The second partial derivative depends on the path, so it doesn't exist at $(0,0)$.

Therefore, f is not of class C^2 .

Exercise 2:

Let f be a function from $\mathbb{R}^2$ to $\mathbb{R}$, defined by:

$$f(x,y) = \frac{x^2 y^2}{x^2 + y^2} \text{ if } (x,y) \neq (0,0) \text{ and } f(0,0) = 0$$

1. Study the continuity of f at $(0,0)$.
2. Calculate the partial derivatives of f at $(0,0)$.
3. Study the differentiability of f at $(0,0)$.
4. Let $(x,y) \neq (0,0)$. Calculate the partial derivatives of f at (x,y) .
5. Is f of class C^1 ?
6. Deduce the answer to (3) from (5).

Correction**1. Continuity at $(0, 0)$:**

Using polar coordinates $x = r \cos \theta$, $y = r \sin \theta$:

$$f(r \cos \theta, r \sin \theta) = \frac{r^4 \cos^2 \theta \sin^2 \theta}{r^2} = r^2 \cos^2 \theta \sin^2 \theta$$

Since $|\cos^2 \theta \sin^2 \theta| \leq 1$, we have:

$$|f(r \cos \theta, r \sin \theta)| \leq r^2 \rightarrow 0 \quad \text{as } r \rightarrow 0$$

Therefore, f is continuous at $(0, 0)$.

2. Partial derivatives at $(0, 0)$:

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0$$

$$\frac{\partial f}{\partial y}(0, 0) = \lim_{k \rightarrow 0} \frac{f(0, k) - f(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{0 - 0}{k} = 0$$

3. Differentiability at $(0, 0)$:

Check the limit:

$$\lim_{(h,k) \rightarrow (0,0)} \frac{f(h, k) - f(0, 0) - 0 \cdot h - 0 \cdot k}{\sqrt{h^2 + k^2}} = \lim_{(h,k) \rightarrow (0,0)} \frac{\frac{h^2 k^2}{h^2 + k^2}}{\sqrt{h^2 + k^2}}$$

In polar coordinates:

$$= \lim_{r \rightarrow 0} \frac{r^4 \cos^2 \theta \sin^2 \theta}{r^2 \cdot r} = \lim_{r \rightarrow 0} r \cos^2 \theta \sin^2 \theta = 0$$

Therefore, f is differentiable at $(0, 0)$.

4. Partial derivatives for $(x, y) \neq (0, 0)$:

$$\frac{\partial f}{\partial x}(x, y) = \frac{2xy^2(x^2 + y^2) - x^2y^2(2x)}{(x^2 + y^2)^2} = \frac{2xy^4}{(x^2 + y^2)^2}$$

$$\frac{\partial f}{\partial y}(x, y) = \frac{2x^2y(x^2 + y^2) - x^2y^2(2y)}{(x^2 + y^2)^2} = \frac{2x^4y}{(x^2 + y^2)^2}$$

5. Class C^1 :

We need to check if the partial derivatives are continuous at $(0, 0)$. In polar coordinates:

$$\left| \frac{\partial f}{\partial x}(r \cos \theta, r \sin \theta) \right| = \left| \frac{2r \cos \theta \cdot r^4 \sin^4 \theta}{r^4} \right| = 2r |\cos \theta \sin^4 \theta| \rightarrow 0$$

Similarly for $\frac{\partial f}{\partial y}$.

Therefore, the partial derivatives are continuous at $(0, 0)$, so f is of class C^1 .

6. Deduction:

Since f is of class C^1 , it is automatically differentiable at $(0, 0)$, which confirms our result from part (3).

Exercise 3:

Let f be a C^1 mapping from $\mathbb{R}^n$ to $\mathbb{R}$. We say that f is positively homogeneous of degree r (where r is a given real number) if and only if $\forall \lambda \in]0, +\infty[, \forall x \in \mathbb{R}^n, f(\lambda x) = \lambda^r f(x)$.

Show that for such a function, the following Euler identity holds:

$$\forall x = (x_1, \dots, x_n) \in \mathbb{R}^n, \quad \sum_{i=1}^n x_i \frac{\partial f}{\partial x_i}(x) = r f(x).$$

Correction

Let $x \in \mathbb{R}^n$ be fixed. Define $\varphi(\lambda) = f(\lambda x)$ for $\lambda > 0$.

By the chain rule:

$$\varphi'(\lambda) = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(\lambda x) \cdot x_i$$

On the other hand, since f is homogeneous of degree r :

$$\varphi(\lambda) = f(\lambda x) = \lambda^r f(x)$$

$$\varphi'(\lambda) = r \lambda^{r-1} f(x)$$

Equating both expressions for $\varphi'(\lambda)$:

$$\sum_{i=1}^n \frac{\partial f}{\partial x_i}(\lambda x) \cdot x_i = r \lambda^{r-1} f(x)$$

Setting $\lambda = 1$:

$$\sum_{i=1}^n \frac{\partial f}{\partial x_i}(x) \cdot x_i = r f(x)$$

This is the Euler identity.

Exercise 4:

Let f be a function from $\mathbb{R}^2$ to $\mathbb{R}$. Let $(a, b) \in \mathbb{R}^2$. Suppose that f is differentiable in a neighborhood of (a, b) . Let S be the surface with equation $z = f(x, y)$.

1. Give the equation of the tangent plane P to S at the point $(a, b, f(a, b))$.
2. Give two direction vectors $\mathbf{u}$ and $\mathbf{v}$ of P and the normal vector to P .

Application:

Let $f(x, y) = \exp(x \ln(x^2 + y^2))$ for $(x, y) \neq (0, 0)$. Give the equation of the tangent plane to the surface $S = \{(x, y, z) \in \mathbb{R}^3 \mid z = f(x, y)\}$ at the point $(1, 0, f(1, 0))$.

Additional question:

Determine the Taylor expansion of f to order 3 at $(0, 1)$ using the uniqueness of a Taylor expansion and certain restrictions of f .

Correction**1. Tangent plane equation:**

The tangent plane to S at $(a, b, f(a, b))$ has equation:

$$z = f(a, b) + \frac{\partial f}{\partial x}(a, b)(x - a) + \frac{\partial f}{\partial y}(a, b)(y - b)$$

2. Direction vectors and normal vector:

Two direction vectors of the tangent plane are:

$$\mathbf{u} = \left(1, 0, \frac{\partial f}{\partial x}(a, b)\right), \quad \mathbf{v} = \left(0, 1, \frac{\partial f}{\partial y}(a, b)\right)$$

The normal vector is:

$$\mathbf{n} = \mathbf{u} \times \mathbf{v} = \left(-\frac{\partial f}{\partial x}(a, b), -\frac{\partial f}{\partial y}(a, b), 1\right)$$

Application:

First, compute $f(1, 0)$:

$$f(1, 0) = \exp(1 \cdot \ln(1^2 + 0^2)) = \exp(\ln 1) = 1$$

Now compute partial derivatives at $(1, 0)$:

Let $u(x, y) = x \ln(x^2 + y^2)$, then $f(x, y) = \exp(u(x, y))$

$$\frac{\partial u}{\partial x} = \ln(x^2 + y^2) + x \cdot \frac{2x}{x^2 + y^2} = \ln(x^2 + y^2) + \frac{2x^2}{x^2 + y^2}$$

$$\frac{\partial u}{\partial y} = x \cdot \frac{2y}{x^2 + y^2} = \frac{2xy}{x^2 + y^2}$$

At $(1, 0)$:

$$\frac{\partial u}{\partial x}(1, 0) = \ln 1 + \frac{2}{1} = 2, \quad \frac{\partial u}{\partial y}(1, 0) = 0$$

Therefore:

$$\frac{\partial f}{\partial x}(1, 0) = f(1, 0) \cdot \frac{\partial u}{\partial x}(1, 0) = 1 \cdot 2 = 2$$

$$\frac{\partial f}{\partial y}(1, 0) = f(1, 0) \cdot \frac{\partial u}{\partial y}(1, 0) = 1 \cdot 0 = 0$$

The tangent plane equation is:

$$z = 1 + 2(x - 1) + 0(y - 0) = 2x - 1$$

Additional question:

Let's find the Taylor expansion of $f(x, y) = \exp(x \ln(x^2 + y^2))$ at $(0, 1)$.

First, simplify $f(x, y)$:

$$f(x, y) = \exp(x \ln(x^2 + y^2)) = (x^2 + y^2)^x$$

Now consider the restriction along $x = 0$:

$$f(0, y) = (0 + y^2)^0 = 1 \quad (\text{by convention } a^0 = 1 \text{ for } a \neq 0)$$

Consider the restriction along $y = 1$:

$$f(x, 1) = (x^2 + 1)^x = \exp(x \ln(x^2 + 1))$$

Let's expand $g(x) = x \ln(x^2 + 1)$ around $x = 0$:

$$\ln(x^2 + 1) = x^2 - \frac{x^4}{2} + O(x^6)$$

$$g(x) = x \cdot \left(x^2 - \frac{x^4}{2} + O(x^6) \right) = x^3 - \frac{x^5}{2} + O(x^7)$$

Therefore:

$$f(x, 1) = \exp \left(x^3 - \frac{x^5}{2} + O(x^7) \right) = 1 + x^3 - \frac{x^5}{2} + O(x^7)$$

Since the Taylor expansion is unique and we need order 3, and from $f(0, y) = 1$, we deduce that the Taylor expansion at $(0, 1)$ to order 3 is:

$$f(x, y) = 1 + x^3 + O(\|(x, y - 1)\|^4)$$

More precisely, the Taylor expansion to order 3 is simply 1, since the first non-zero term appears at order 3.

Exercise 5:

Find the extrema of the following functions:

1. $f(x, y) = x^3 + 3x^2y - 15x - 12y$
2. $f(x, y) = -2(x - y)^2 + x^4 + y^4$

Correction

1. For $f(x, y) = x^3 + 3x^2y - 15x - 12y$:

Find critical points:

$$\frac{\partial f}{\partial x} = 3x^2 + 6xy - 15 = 0 \quad (1)$$

$$\frac{\partial f}{\partial y} = 3x^2 - 12 = 0 \quad (2)$$

From (2): $3x^2 - 12 = 0 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2$

Case 1: $x = 2$ Substitute in (1): $3(4) + 6(2)y - 15 = 12 + 12y - 15 = 12y - 3 = 0 \Rightarrow y = \frac{1}{4}$

Case 2: $x = -2$ Substitute in (1): $3(4) + 6(-2)y - 15 = 12 - 12y - 15 = -12y - 3 = 0 \Rightarrow y = -\frac{1}{4}$

Critical points: $(2, \frac{1}{4})$ and $(-2, -\frac{1}{4})$

Now compute second derivatives:

$$\frac{\partial^2 f}{\partial x^2} = 6x + 6y, \quad \frac{\partial^2 f}{\partial y^2} = 0, \quad \frac{\partial^2 f}{\partial x \partial y} = 6x$$

Hessian matrix: $H = \begin{pmatrix} 6x + 6y & 6x \\ 6x & 0 \end{pmatrix}$

At $(2, \frac{1}{4})$: $H = \begin{pmatrix} 12 + 1.5 & 12 \\ 12 & 0 \end{pmatrix} = \begin{pmatrix} 13.5 & 12 \\ 12 & 0 \end{pmatrix}$ Determinant: $\det(H) = 13.5 \cdot 0 - 12 \cdot 12 = -144 < 0$, so saddle point.

At $(-2, -\frac{1}{4})$: $H = \begin{pmatrix} -12 - 1.5 & -12 \\ -12 & 0 \end{pmatrix} = \begin{pmatrix} -13.5 & -12 \\ -12 & 0 \end{pmatrix}$ Determinant: $\det(H) = (-13.5) \cdot 0 - (-12) \cdot (-12) = -144 < 0$, so saddle point.

Both critical points are saddle points.

2. For $f(x, y) = -2(x - y)^2 + x^4 + y^4$:

Find critical points:

$$\frac{\partial f}{\partial x} = -4(x - y) + 4x^3 = 0 \quad (1)$$

$$\frac{\partial f}{\partial y} = 4(x - y) + 4y^3 = 0 \quad (2)$$

Add (1) and (2): $4x^3 + 4y^3 = 0 \Rightarrow x^3 + y^3 = 0 \Rightarrow (x + y)(x^2 - xy + y^2) = 0$

Since $x^2 - xy + y^2 > 0$ for $(x, y) \neq (0, 0)$, we have $x + y = 0 \Rightarrow y = -x$

Substitute in (1): $-4(x - (-x)) + 4x^3 = -8x + 4x^3 = 0 \Rightarrow 4x(x^2 - 2) = 0$

So $x = 0$ or $x = \pm\sqrt{2}$

Critical points: $(0, 0)$, $(\sqrt{2}, -\sqrt{2})$, $(-\sqrt{2}, \sqrt{2})$

Second derivatives:

$$\frac{\partial^2 f}{\partial x^2} = -4 + 12x^2, \quad \frac{\partial^2 f}{\partial y^2} = -4 + 12y^2, \quad \frac{\partial^2 f}{\partial x \partial y} = 4$$

At $(0, 0)$: $H = \begin{pmatrix} -4 & 4 \\ 4 & -4 \end{pmatrix}$ Determinant: $\det(H) = 16 - 16 = 0$, so test fails.

Let's analyze $f(x, y)$ near $(0, 0)$: Along $y = x$: $f(x, x) = -2(0)^2 + x^4 + x^4 = 2x^4 \geq 0$

Along $y = -x$: $f(x, -x) = -2(2x)^2 + x^4 + x^4 = -8x^2 + 2x^4$

For small x , $-8x^2 + 2x^4 < 0$, so $(0, 0)$ is a saddle point.

At $(\sqrt{2}, -\sqrt{2})$: $H = \begin{pmatrix} -4 + 12(2) & 4 \\ 4 & -4 + 12(2) \end{pmatrix} = \begin{pmatrix} 20 & 4 \\ 4 & 20 \end{pmatrix}$ Determinant:

$\det(H) = 400 - 16 = 384 > 0$ and $\frac{\partial^2 f}{\partial x^2} = 20 > 0$, so local minimum.

At $(-\sqrt{2}, \sqrt{2})$: $H = \begin{pmatrix} -4 + 12(2) & 4 \\ 4 & -4 + 12(2) \end{pmatrix} = \begin{pmatrix} 20 & 4 \\ 4 & 20 \end{pmatrix}$ Same as above, so local minimum.

Value at minima: $f(\pm\sqrt{2}, \mp\sqrt{2}) = -2(2\sqrt{2})^2 + 4 + 4 = -16 + 8 = -8$

Exercise 6:

Find the minimum of $f(x, y) = \sqrt{x^2 + (y - a)^2} + \sqrt{(x - a)^2 + y^2}$, where a is a given real number.

Correction

This function represents the sum of distances from point (x, y) to points $(0, a)$ and $(a, 0)$.

By the triangle inequality, the shortest path between two points is a straight line. The minimum occurs when (x, y) lies on the line segment connecting $(0, a)$ and $(a, 0)$.

The line through $(0, a)$ and $(a, 0)$ has equation $x + y = a$.

On this line, the distance is simply the distance between $(0, a)$ and $(a, 0)$, which is:

$$\sqrt{(a-0)^2 + (0-a)^2} = \sqrt{a^2 + a^2} = a\sqrt{2}$$

So the minimum value is $a\sqrt{2}$, achieved for any point (x, y) on the line segment $x + y = a$ with $0 \leq x \leq a$.

To verify analytically, we can use symmetry. The function is symmetric in x and y .

Try $y = a - x$:

$$f(x, a-x) = \sqrt{2x^2} + \sqrt{2(x-a)^2} = \sqrt{2}(|x| + |x-a|)$$

For $0 \leq x \leq a$, this becomes $\sqrt{2}(x + a - x) = a\sqrt{2}$, which is constant.

Therefore, the minimum value is $a\sqrt{2}$.

Exercise 7:

Solve the following partial differential equations:

1. $2\frac{\partial f}{\partial x} - \frac{\partial f}{\partial y} = 0$ by setting $u = x + y$ and $v = x + 2y$.
2. $x\frac{\partial f}{\partial y} - y\frac{\partial f}{\partial x} = 0$ on $\mathbb{R}^2 \setminus \{(0, 0)\}$ by using polar coordinates.
3. $x^2\frac{\partial^2 f}{\partial x^2} + 2xy\frac{\partial^2 f}{\partial x\partial y} + y^2\frac{\partial^2 f}{\partial y^2} = 0$ on $]0, +\infty[\times \mathbb{R}$ by setting $x = u$ and $y = uv$.

Correction

1. $2\frac{\partial f}{\partial x} - \frac{\partial f}{\partial y} = 0$ **with** $u = x + y$, $v = x + 2y$:

Compute the Jacobian:

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} = 2 - 1 = 1$$

Now express partial derivatives in new coordinates:

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial x} = \frac{\partial f}{\partial u} + \frac{\partial f}{\partial v}$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial y} = \frac{\partial f}{\partial u} + 2\frac{\partial f}{\partial v}$$

Substitute in the PDE:

$$2 \left(\frac{\partial f}{\partial u} + \frac{\partial f}{\partial v} \right) - \left(\frac{\partial f}{\partial u} + 2 \frac{\partial f}{\partial v} \right) = 2 \frac{\partial f}{\partial u} + 2 \frac{\partial f}{\partial v} - \frac{\partial f}{\partial u} - 2 \frac{\partial f}{\partial v} = \frac{\partial f}{\partial u} = 0$$

So $\frac{\partial f}{\partial u} = 0$, which means f depends only on v :

$$f(x, y) = g(v) = g(x + 2y)$$

where g is an arbitrary differentiable function.

2. $x \frac{\partial f}{\partial y} - y \frac{\partial f}{\partial x} = 0$ using polar coordinates:

Let $x = r \cos \theta$, $y = r \sin \theta$

Compute partial derivatives:

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial f}{\partial \theta} \frac{\partial \theta}{\partial x} = \frac{\partial f}{\partial r} \cos \theta - \frac{\partial f}{\partial \theta} \frac{\sin \theta}{r}$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial f}{\partial \theta} \frac{\partial \theta}{\partial y} = \frac{\partial f}{\partial r} \sin \theta + \frac{\partial f}{\partial \theta} \frac{\cos \theta}{r}$$

Substitute in the PDE:

$$x \left(\frac{\partial f}{\partial r} \sin \theta + \frac{\partial f}{\partial \theta} \frac{\cos \theta}{r} \right) - y \left(\frac{\partial f}{\partial r} \cos \theta - \frac{\partial f}{\partial \theta} \frac{\sin \theta}{r} \right) = 0$$

Substitute $x = r \cos \theta$, $y = r \sin \theta$:

$$r \cos \theta \left(\frac{\partial f}{\partial r} \sin \theta + \frac{\partial f}{\partial \theta} \frac{\cos \theta}{r} \right) - r \sin \theta \left(\frac{\partial f}{\partial r} \cos \theta - \frac{\partial f}{\partial \theta} \frac{\sin \theta}{r} \right) = 0$$

Simplify:

$$r \frac{\partial f}{\partial r} (\cos \theta \sin \theta - \sin \theta \cos \theta) + \frac{\partial f}{\partial \theta} (\cos^2 \theta + \sin^2 \theta) = \frac{\partial f}{\partial \theta} = 0$$

So $\frac{\partial f}{\partial \theta} = 0$, which means f depends only on r :

$$f(x, y) = g(r) = g(\sqrt{x^2 + y^2})$$

where g is an arbitrary differentiable function.

3. $x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = 0$ with $x = u$, $y = uv$:

Compute partial derivatives:

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial x} = \frac{\partial f}{\partial u} - \frac{y}{x^2} \frac{\partial f}{\partial v} = \frac{\partial f}{\partial u} - \frac{v}{u} \frac{\partial f}{\partial v}$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial y} = 0 + \frac{1}{x} \frac{\partial f}{\partial v} = \frac{1}{u} \frac{\partial f}{\partial v}$$

Now compute second derivatives:

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial u} \left(\frac{\partial f}{\partial u} - \frac{v}{u} \frac{\partial f}{\partial v} \right) - \frac{v}{u} \frac{\partial}{\partial v} \left(\frac{\partial f}{\partial u} - \frac{v}{u} \frac{\partial f}{\partial v} \right)$$

$$\begin{aligned}
&= \frac{\partial^2 f}{\partial u^2} - \frac{v}{u} \frac{\partial^2 f}{\partial u \partial v} + \frac{v}{u^2} \frac{\partial f}{\partial v} - \frac{v}{u^2} \frac{\partial f}{\partial v} - \frac{v}{u} \left(\frac{\partial^2 f}{\partial u \partial v} - \frac{v}{u} \frac{\partial^2 f}{\partial v^2} \right) \\
&= \frac{\partial^2 f}{\partial u^2} - \frac{2v}{u} \frac{\partial^2 f}{\partial u \partial v} + \frac{v^2}{u^2} \frac{\partial^2 f}{\partial v^2}
\end{aligned}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial v} \left(\frac{1}{u} \frac{\partial f}{\partial v} \right) \frac{\partial v}{\partial y} = \frac{1}{u^2} \frac{\partial^2 f}{\partial v^2}$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial u} \left(\frac{1}{u} \frac{\partial f}{\partial v} \right) - \frac{v}{u} \frac{\partial}{\partial v} \left(\frac{1}{u} \frac{\partial f}{\partial v} \right) = -\frac{1}{u^2} \frac{\partial f}{\partial v} + \frac{1}{u} \frac{\partial^2 f}{\partial u \partial v} - \frac{v}{u^2} \frac{\partial^2 f}{\partial v^2}$$

Now substitute in the PDE:

$$\begin{aligned}
&x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} \\
&= u^2 \left(\frac{\partial^2 f}{\partial u^2} - \frac{2v}{u} \frac{\partial^2 f}{\partial u \partial v} + \frac{v^2}{u^2} \frac{\partial^2 f}{\partial v^2} \right) + 2u(uv) \left(-\frac{1}{u^2} \frac{\partial f}{\partial v} + \frac{1}{u} \frac{\partial^2 f}{\partial u \partial v} - \frac{v}{u^2} \frac{\partial^2 f}{\partial v^2} \right) + (uv)^2 \left(\frac{1}{u^2} \frac{\partial^2 f}{\partial v^2} \right)
\end{aligned}$$

Simplify:

$$\begin{aligned}
&= u^2 \frac{\partial^2 f}{\partial u^2} - 2uv \frac{\partial^2 f}{\partial u \partial v} + v^2 \frac{\partial^2 f}{\partial v^2} - 2v \frac{\partial f}{\partial v} + 2uv \frac{\partial^2 f}{\partial u \partial v} - 2v^2 \frac{\partial^2 f}{\partial v^2} + v^2 \frac{\partial^2 f}{\partial v^2} \\
&= u^2 \frac{\partial^2 f}{\partial u^2} - 2v \frac{\partial f}{\partial v} = 0
\end{aligned}$$

So we get:

$$u^2 \frac{\partial^2 f}{\partial u^2} - 2v \frac{\partial f}{\partial v} = 0$$

This is a separable PDE. Let's solve it:

The equation is: $u^2 \frac{\partial^2 f}{\partial u^2} = 2v \frac{\partial f}{\partial v}$

Assume separation of variables: $f(u, v) = U(u)V(v)$

Then: $u^2 U''(u)V(v) = 2vU(u)V'(v)$

Divide by $U(u)V(v)$: $\frac{u^2 U''(u)}{U(u)} = \frac{2v V'(v)}{V(v)} = \lambda$ (constant)

So we have two ODEs:

$$u^2 U''(u) - \lambda U(u) = 0$$

$$2v V'(v) - \lambda V(v) = 0$$

The general solution is a linear combination of separated solutions.

Alternatively, we can see that the PDE in (u, v) coordinates suggests that f depends on u and v in a specific way. The equation $u^2 \frac{\partial^2 f}{\partial u^2} = 2v \frac{\partial f}{\partial v}$ is an Euler-type equation. The general solution is:

$$f(x, y) = A\left(\frac{y}{x}\right) + xB\left(\frac{y}{x}\right)$$

where A and B are arbitrary twice differentiable functions.

Exercise 8:

Let f be a real-valued function of a real variable defined in a neighborhood of a point a such that $f''(a)$ exists.

1. Let $\varepsilon > 0$ be a given real number. Show that for t sufficiently close to a , we have the inequality:

$$|f'(t) - f'(a) - (t - a)f''(a)| < \varepsilon|t - a|$$

2. Define $H(x) = f(x) - (x - a)f'(a) - \frac{(x - a)^2}{2}f''(a)$.

Deduce a bound for $H(y) - H(x)$ when x and y are sufficiently close to a .

3. Define:

$$F(x, y) = \frac{f(y) - f(x)}{y - x} \text{ for } y \neq x \text{ and } F(x, x) = f'(x)$$

Show that F is differentiable at the point (a, a) and that:

$$dF_{(a,a)} = \frac{1}{2}f''(a)(dx + dy)$$

Correction

1. Proof of the inequality:

Since $f''(a)$ exists, we have:

$$\lim_{t \rightarrow a} \frac{f'(t) - f'(a)}{t - a} = f''(a)$$

This means that for any $\varepsilon > 0$, there exists $\delta > 0$ such that for all t with $0 < |t - a| < \delta$, we have:

$$\left| \frac{f'(t) - f'(a)}{t - a} - f''(a) \right| < \varepsilon$$

Multiplying both sides by $|t - a|$:

$$|f'(t) - f'(a) - (t - a)f''(a)| < \varepsilon|t - a|$$

2. Bound for $H(y) - H(x)$:

We have:

$$H(x) = f(x) - (x - a)f'(a) - \frac{(x - a)^2}{2}f''(a)$$

Differentiate:

$$H'(x) = f'(x) - f'(a) - (x - a)f''(a)$$

From part (1), for x sufficiently close to a , we have:

$$|H'(x)| < \varepsilon|x - a|$$

Now, for x and y sufficiently close to a , by the Mean Value Theorem, there exists c between x and y such that:

$$H(y) - H(x) = H'(c)(y - x)$$

Therefore:

$$|H(y) - H(x)| = |H'(c)||y - x| < \varepsilon|c - a||y - x|$$

Since c is between x and y , and x, y are close to a , we have $|c - a| \leq \max(|x - a|, |y - a|)$. So:

$$|H(y) - H(x)| < \varepsilon \max(|x - a|, |y - a|)|y - x|$$

3. Differentiability of F at (a, a) :

We need to show that:

$$\lim_{(x,y) \rightarrow (a,a)} \frac{F(x,y) - F(a,a) - \frac{1}{2}f''(a)((x-a) + (y-a))}{\sqrt{(x-a)^2 + (y-a)^2}} = 0$$

Note that $F(a, a) = f'(a)$.

Consider the numerator:

$$F(x, y) - F(a, a) - \frac{1}{2}f''(a)((x-a) + (y-a)) = \frac{f(y) - f(x)}{y - x} - f'(a) - \frac{1}{2}f''(a)((x-a) + (y-a))$$

Using the definition of H :

$$f(x) = H(x) + (x - a)f'(a) + \frac{(x - a)^2}{2}f''(a)$$

$$f(y) = H(y) + (y - a)f'(a) + \frac{(y - a)^2}{2}f''(a)$$

So:

$$f(y) - f(x) = H(y) - H(x) + (y - x)f'(a) + \frac{1}{2}f''(a)[(y - a)^2 - (x - a)^2]$$

Divide by $y - x$:

$$\frac{f(y) - f(x)}{y - x} = \frac{H(y) - H(x)}{y - x} + f'(a) + \frac{1}{2}f''(a) \frac{(y - a)^2 - (x - a)^2}{y - x}$$

Note that:

$$\frac{(y - a)^2 - (x - a)^2}{y - x} = \frac{(y - a - (x - a))(y - a + (x - a))}{y - x} = \frac{(y - x)(y + x - 2a)}{y - x} = y + x - 2a$$

So:

$$\frac{f(y) - f(x)}{y - x} = \frac{H(y) - H(x)}{y - x} + f'(a) + \frac{1}{2}f''(a)(y + x - 2a)$$

Therefore:

$$\begin{aligned} F(x, y) - F(a, a) - \frac{1}{2}f''(a)((x-a) + (y-a)) &= \frac{H(y) - H(x)}{y - x} + \frac{1}{2}f''(a)(y + x - 2a - (x - a) - (y - a)) \\ &= \frac{H(y) - H(x)}{y - x} + \frac{1}{2}f''(a)(y + x - 2a - x + a - y + a) = \frac{H(y) - H(x)}{y - x} \end{aligned}$$

So we need to show:

$$\lim_{(x,y) \rightarrow (a,a)} \frac{\frac{H(y)-H(x)}{y-x}}{\sqrt{(x-a)^2 + (y-a)^2}} = 0$$

From part (2), we have:

$$|H(y) - H(x)| < \varepsilon \max(|x - a|, |y - a|)|y - x|$$

So:

$$\left| \frac{H(y) - H(x)}{y - x} \right| < \varepsilon \max(|x - a|, |y - a|)$$

Therefore:

$$\left| \frac{\frac{H(y)-H(x)}{y-x}}{\sqrt{(x-a)^2 + (y-a)^2}} \right| < \varepsilon \frac{\max(|x-a|, |y-a|)}{\sqrt{(x-a)^2 + (y-a)^2}} \leq \varepsilon$$

Since ε is arbitrary, the limit is 0.

This proves that F is differentiable at (a, a) and:

$$dF_{(a,a)}(h, k) = \frac{1}{2} f''(a)(h + k)$$

or in differential form:

$$dF_{(a,a)} = \frac{1}{2} f''(a)(dx + dy)$$